GABRIELLA'S MATH LAB · DECIMALS WORKBOOK

SECTION 3

Rounding Decimals

Find the closer number: round to the nearest whole, tenth, or hundredth.

SKILL
One new idea at a time.

SHOW HOW
Watch a problem solved step by step.

TRY ONE
Finish a problem with help.

YOUR TURN
Solve problems on your own.

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Name _____ Started _____

Georgia K-12 Mathematics Standards 5.NR.4.3 (round decimals to the hundredths place). Rounding to the nearest whole number is review.

3.1 Rounding on a Number Line

Goal: I can use a number line to find the closer whole number.

SHOW HOW

Round 2.7 to the nearest whole number.



1. 2.7 is between the whole numbers **2** and **3**. Put them at the ends of the line.
2. Find the **halfway** point: **2.5** (the tall middle tick).
3. 2.7 is past halfway, so it is **closer to 3**. 2.7 rounds to **3**.
4. **Exactly halfway?** 2.5 rounds **up** to 3. That is the rule everyone uses.

Why it works: rounding means "which number is closer?" The halfway point splits the line: left half goes down, right half (and halfway) goes up.

LET'S TRY ONE

Round 6.3 to the nearest whole number.



6.3 is between _____ and _____. Halfway is _____

6.3 is closer to _____. So 6.3 rounds to _____

YOUR TURN — SKILL 3.1 · ROUND TO THE NEAREST WHOLE NUMBER

1



Rounds to _____

2



Rounds to _____

3



Rounds to _____

4



Rounds to _____

5



Rounds to _____

6



Rounds to _____

7 Round **3.9** to the nearest whole number. Draw a number line if it helps.

Rounds to _____

8 Round **15.5** to the nearest whole number. Draw a number line if it helps.

Rounds to _____

3.2 Rounding to the Nearest Whole Number

Goal: I can round using the rounding digit and the digit to its right.

SHOW HOW

Round 4.735 to the nearest whole number.

Ones 1		Tenths 0.1	Hundredths 0.01	Thousandths 0.001
4	.	7	3	5

1. **Find the rounding digit.** Nearest whole number means the **ones** place: the **4**.
2. **Look at the digit to its right:** the **7** in the tenths place.
3. **5 or more? Round up.** 7 is 5 or more, so the 4 goes up to **5**. (0 to 4? Keep the rounding digit.)
4. **Drop the digits after the rounding place: 4.735 → 5**

Why it works: the digit to the right tells you which half you are in. 4.7 is past the halfway point 4.5, so 4.735 is closer to 5. Only that one digit matters.

LET'S TRY ONE

Round 8.29 to the nearest whole number.

Rounding digit (ones): _____ Digit to its right: _____

Round up or keep? _____ 8.29 rounds to _____

YOUR TURN — SKILL 3.2 · ROUND TO THE NEAREST WHOLE NUMBER

1 **3.4**

Rounds to _____

2 **8.6**

Rounds to _____

3 **12.29**

Rounds to _____

4 **0.81**

Rounds to _____

5 **6.499** (look at only one digit!)

Rounds to _____

6 **19.5**

Rounds to _____

7 **24.73**

Rounds to _____

8 **9.62** (9 goes up to 10)

Rounds to _____

3.3 Rounding to the Nearest Tenth

Goal: I can round a decimal to the nearest tenth.

SHOW HOW

Round 4.76 to the nearest tenth.



1. **Rounding digit:** the **tenths** digit, **7**. So 4.76 is between **4.7** and **4.8**.
2. **Digit to its right:** the hundredths digit, **6**. 6 is 5 or more, so **round up**: 7 becomes 8.
3. Drop the digits after the tenths place: **4.76 → 4.8**. The number line agrees: 4.76 is past halfway (4.75).

Why it works: it is the same rule as rounding to a whole number. Only the rounding place changes. When a 9 rounds up, carry it: 2.96 → 3.0.

LET'S TRY ONE

Round 5.32 to the nearest tenth.

Rounding digit (tenths): _____ Digit to its right: _____

Round up or keep? _____ 5.32 rounds to _____

YOUR TURN — SKILL 3.3 · ROUND TO THE NEAREST TENTH

1 3.42

Rounds to _____

2 0.68

Rounds to _____

3 6.15

Rounds to _____

4 12.349

Rounds to _____

5 0.451

Rounds to _____

6 8.04

Rounds to _____

7 7.96

Rounds to _____

8 19.95

Rounds to _____

3.4 Rounding to the Nearest Hundredth

Goal: I can round a decimal to the nearest hundredth.

SHOW HOW

Round 4.735 to the nearest hundredth.



1. **Rounding digit:** the **hundredths** digit, **3**. So 4.735 is between **4.73** and **4.74**.
2. **Digit to its right:** the thousandths digit, **5**. 5 means **round up**: 3 becomes 4.
3. Drop the digit after the hundredths place: **4.735 → 4.74**. On the line, 4.735 is exactly halfway, and halfway rounds up.

Why it works: same two questions every time: **Where is the rounding digit?** and **Is the next digit 5 or more?** Keep the zeros you need: 3.995 → 4.00.

LET'S TRY ONE

Round 2.348 to the nearest hundredth.

Rounding digit (hundredths): _____ Digit to its right: _____

Round up or keep? _____ 2.348 rounds to _____

YOUR TURN — SKILL 3.4 · ROUND TO THE NEAREST HUNDREDTH

1 2.341

Rounds to _____

2 0.784

Rounds to _____

3 5.102

Rounds to _____

4 8.065

Rounds to _____

5 7.123

Rounds to _____

6 0.009

Rounds to _____

7 9.507

Rounds to _____

8 3.995

Rounds to _____

3.5 Rounding Word Problems

Goal: I can find the number and the rounding place in a story, then round.

SHOW HOW

Maya ran 3.68 miles. About how far did she run, to the nearest tenth of a mile?

1. **Find the number:** 3.68. **Find the rounding place:** "nearest tenth" → the **tenths** digit, 6.
2. **Digit to its right:** 8. 8 is 5 or more, so round up: 6 becomes 7.
3. **Answer with units:** Maya ran about **3.7 miles**.

Clue words: "about", "nearly", "estimate", "to the nearest …". Money to the nearest **dollar** = nearest whole number; to the nearest **dime** = nearest tenth; to the nearest **cent** = nearest hundredth.

LET'S TRY ONE

A book costs $12.45. What is the price to the nearest dollar?

Number: _____ Rounding place: _____

Rounding digit: _____ Digit to its right: _____ About $_____

YOUR TURN — SKILL 3.5 · ROUND TO ANSWER THE QUESTION

1 A pumpkin weighs **4.8 pounds**. About how many pounds is that, to the nearest whole pound?

About _____ pounds

2 A jug holds **2.35 liters**. Round to the nearest tenth of a liter.

About _____ liters

3 A toy costs **$7.62**. What is the price to the nearest dollar?

About $ _____

4 A snail crawled **0.375 meter**. Round to the nearest hundredth of a meter.

About _____ m

5 Leo swam a lap in **9.86 seconds**. Round his time to the nearest tenth of a second.

About _____ seconds

6 A pencil box is **15.49 cm** long. Round to the nearest whole centimeter.

About _____ cm

7 A trail is **3.96 miles** long. Round to the nearest tenth of a mile.

About _____ miles

8 A science scale shows **25.555 grams**. Round to the nearest hundredth of a gram.

About _____ g

Section 3 Review

Mixed practice from Skills 3.1 to 3.5. No hints: show what you know.

YOUR TURN

REVIEW · PROBLEMS 1–8

1 Round **5.6** to the nearest whole number.

Rounds to _____

2 Round **13.45** to the nearest whole number.

Rounds to _____

3 Round **0.74** to the nearest tenth.

Rounds to _____

4 Round **6.98** to the nearest tenth.

Rounds to _____

5 Round **1.234** to the nearest hundredth.

Rounds to _____

6 Round **4.505** to the nearest hundredth.

Rounds to _____

7 A backpack weighs **2.65 kg**. Round to the nearest tenth of a kilogram.

About _____ kg

8 Write a number with **tenths** that rounds to **3** when you round to the nearest whole number.

Number _____

Parent Answer Key

When rounding to tenths, 8.0 and 8 are both correct (8.0 shows the rounding place). “Try” is the Let's Try One problem.

3.1 Rounding on a Number Line

- **Try** **6** 6.3 is between 6 and 7. Halfway is 6.5. 6.3 is before halfway, so it is closer to 6.
- **1** **4** 4.2 is between 4 and 5. Halfway is 4.5. 4.2 is before halfway, so it rounds to 4.
- **2** **2** 1.8 is between 1 and 2. Halfway is 1.5. 1.8 is past halfway, so it rounds to 2.
- **3** **8** 7.6 is between 7 and 8. Halfway is 7.5. 7.6 is past halfway, so it rounds to 8.
- **4** **0** 0.4 is between 0 and 1. Halfway is 0.5. 0.4 is before halfway, so it rounds to 0.
- **5** **10** 9.5 is between 9 and 10. Halfway is 9.5. 9.5 is exactly halfway, and halfway rounds up, so it rounds to 10.
- **6** **12** 12.1 is between 12 and 13. Halfway is 12.5. 12.1 is before halfway, so it rounds to 12.
- **7** **4** Rounding digit: 3 (ones). Digit to its right: 9, 5 or more, so round up → 4.
- **8** **16** Rounding digit: 5 (ones). Digit to its right: 5, 5 or more, so round up → 16.

3.2 Rounding to the Nearest Whole Number

- **Try** **8** Rounding digit: 8 (ones). Digit to its right: 2, less than 5, so keep → 8.
- **1** **3** Rounding digit: 3 (ones). Digit to its right: 4, less than 5, so keep → 3.
- **2** **9** Rounding digit: 8 (ones). Digit to its right: 6, 5 or more, so round up → 9.
- **3** **12** Rounding digit: 2 (ones). Digit to its right: 2, less than 5, so keep → 12.
- **4** **1** Rounding digit: 0 (ones). Digit to its right: 8, 5 or more, so round up → 1.
- **5** **6** Rounding digit: 6 (ones). Digit to its right: 4, less than 5, so keep → 6.
- **6** **20** Rounding digit: 9 (ones). Digit to its right: 5, 5 or more, so round up → 20.
- **7** **25** Rounding digit: 4 (ones). Digit to its right: 7, 5 or more, so round up → 25.
- **8** **10** Rounding digit: 9 (ones). Digit to its right: 6, 5 or more, so round up → 10.

3.3 Rounding to the Nearest Tenth

- **Try** **5.3** Rounding digit: 3 (tenths). Digit to its right: 2, less than 5, so keep → 5.3.
- **1** **3.4** Rounding digit: 4 (tenths). Digit to its right: 2, less than 5, so keep → 3.4.
- **2** **0.7** Rounding digit: 6 (tenths). Digit to its right: 8, 5 or more, so round up → 0.7.
- **3** **6.2** Rounding digit: 1 (tenths). Digit to its right: 5, 5 or more, so round up → 6.2.
- **4** **12.3** Rounding digit: 3 (tenths). Digit to its right: 4, less than 5, so keep → 12.3.
- **5** **0.5** Rounding digit: 4 (tenths). Digit to its right: 5, 5 or more, so round up → 0.5.
- **6** **8.0** Rounding digit: 0 (tenths). Digit to its right: 4, less than 5, so keep → 8.0.
- **7** **8.0** Rounding digit: 9 (tenths). Digit to its right: 6, round up. 9 tenths + 1 tenth = 10 tenths = 1 whole, so 7.96 → 8.0 (also written 8).
- **8** **20.0** Rounding digit: 9 (tenths). Digit to its right: 5, round up. 19.9 + 0.1 = 20.0, so 19.95 → 20.0 (also written 20).

3.4 Rounding to the Nearest Hundredth

Try **2.35** Rounding digit: 4 (hundredths). Digit to its right: 8, 5 or more, so round up → 2.35.

1 **2.34** Rounding digit: 4 (hundredths). Digit to its right: 1, less than 5, so keep → 2.34.

2 **0.78** Rounding digit: 8 (hundredths). Digit to its right: 4, less than 5, so keep → 0.78.

3 **5.10** Rounding digit: 0 (hundredths). Digit to its right: 2, less than 5, so keep → 5.10.

4 **8.07** Rounding digit: 6 (hundredths). Digit to its right: 5, 5 or more, so round up → 8.07.

5 **7.12** Rounding digit: 2 (hundredths). Digit to its right: 3, less than 5, so keep → 7.12.

6 **0.01** Rounding digit: 0 (hundredths). Digit to its right: 9, 5 or more, so round up → 0.01.

7 **9.51** Rounding digit: 0 (hundredths). Digit to its right: 7, 5 or more, so round up → 9.51.

8 **4.00** Rounding digit: 9 (hundredths). Digit to its right: 5, round up. 3.99 + 0.01 = 4.00, so 3.995 → 4.00 (also written 4).

3.5 Rounding Word Problems

Try **12** Number: 12.45; nearest dollar = ones. Rounding digit 2, digit to its right 4 (less than 5), keep: about $12.

1 **5** Rounding digit: 4 (ones). Digit to its right: 8, 5 or more, so round up → 5.

2 **2.4** Rounding digit: 3 (tenths). Digit to its right: 5, 5 or more, so round up → 2.4.

3 **8** Rounding digit: 7 (ones). Digit to its right: 6, 5 or more, so round up → 8.

4 **0.38** Rounding digit: 7 (hundredths). Digit to its right: 5, 5 or more, so round up → 0.38.

5 **9.9** Rounding digit: 8 (tenths). Digit to its right: 6, 5 or more, so round up → 9.9.

6 **15** Rounding digit: 5 (ones). Digit to its right: 4, less than 5, so keep → 15.

7 **4.0** Rounding digit: 9 (tenths). Digit to its right: 6, 5 or more, so round up → 4.0.

8 **25.56** Rounding digit: 5 (hundredths). Digit to its right: 5, 5 or more, so round up → 25.56.

Review Section 3 Review

1 **6** Rounding digit: 5 (ones). Digit to its right: 6, 5 or more, so round up → 6.

2 **13** Rounding digit: 3 (ones). Digit to its right: 4, less than 5, so keep → 13.

3 **0.7** Rounding digit: 7 (tenths). Digit to its right: 4, less than 5, so keep → 0.7.

4 **7.0** Rounding digit: 9 (tenths). Digit to its right: 8, 5 or more, so round up → 7.0.

5 **1.23** Rounding digit: 3 (hundredths). Digit to its right: 4, less than 5, so keep → 1.23.

6 **4.51** Rounding digit: 0 (hundredths). Digit to its right: 5, 5 or more, so round up → 4.51.

7 **2.7** Rounding digit: 6 (tenths). Digit to its right: 5, 5 or more, so round up → 2.7.

8 **2.5, 2.6, 2.7, 2.8, 2.9, 3.0, 3.1, 3.2, 3.3, 3.4 (any one of these)** To round to 3, a number with tenths must be at least 2.5 and less than 3.5. Any of these is correct.